

Element E: Application of STEM Principles and Practices

STEM Principle for High-Priority Design Requirement #1-Moves based on desired amount of sunlight

STEM Concept:Technology and science

Describe how the STEM concept applies directly to the specifics of your design: We have to use technology to figure out how to program the device; to move based on the needed amount of sunlight, and the science is needed to figure out how much sunlight is needed.

STEM Principle for High-Priority Design Requirement #2-Programmable for time of day and total time in the sun

STEM Concept:Technology

Describe how STEM concept applies directly to the specifics of your design: The programming portion needs to use technology to be used. (batteries, programmable device, etc.)

STEM Principle for High-Priority Design Requirement #3-Must use a battery

STEM Concept: Technology

Describe how STEM concept applies directly to the specifics of your design: The battery is technology

STEM Principle for High-Priority Design Requirement #4- Be a 10 by 20 platform

STEM Concept:Math

Describe how STEM concept applies directly to the specifics of your design: This is math because you have to measure the platform and cut it to the proper length.

STEM Principle for High-Priority Design Requirement #5-Hold up to 10 lbs

STEM Concept:Math and Engineering

Describe how STEM concept applies directly to the specifics of your design: You have to create the design to hold up to 10 lbs, without breaking.